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Intra-particle diffusion limitation for steam methane reforming over a Ni-based catalyst

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ABSTRACT

This work aims to evaluate the diffusion limitation for the steam methane reforming process over spherical catalyst particles (Ni/ α -Al₂O₃). Numerical calculations based on a CFD model were conducted to determine the intrinsic and actual reaction rates. The CFD model was validated using experimental results, and good agreement was achieved. The effectiveness factor for the reaction rates was used to express the effect of diffusion limitation within the porous catalyst particle. The effectiveness factor was determined for a wide range of operational parameters, including a temperature range of 800–1300 K and a steam-to-methane ratio of 1–5. The analysis was performed for spherical catalyst particles with diameters up to 3.42 mm, and the effectiveness factor as a function of catalyst particle diameter, temperature, and steam-to-methane ratio was established. The results show that the effectiveness factor is inversely proportional to the diameter of the catalyst particle and that an increase in reaction temperature and steam-to-methane ratio leads to an increase in diffusion limitation. The limitations of the analysis are discussed and justified. The final expression for the effectiveness factor can be used in a pseudo-homogeneous steam methane reformer model where the effectiveness factor varies depending on the position in the packed bed.

1. Introduction

The energy transition from hydrocarbon fuels to carbon-free fuels is the reality nowadays. There are different scenarios for this transition but the consensus in many countries is to end carbon dioxide emissions by the middle of the 21st century [1,2]. Hydrogen is a promising energy carrier that can be considered as an alternative to hydrocarbon fuels. The different approaches are utilized for hydrogen production: hydrocarbon fuel reforming without CO₂ capture (grey hydrogen), hydrocarbon fuel reforming with CO₂ capture (blue hydrogen), water electrolysis with renewable energy sources (green hydrogen), etc [3]. In spite of significant progress in the development of ways to produce green hydrogen, grey hydrogen will be dominant in the coming decades. Based on S&P Global Commodity Insight (formerly IHS Markit) [4], in 2030 the annual hydrogen production will be in these proportions: 88% – grey hydrogen, 9% – blue hydrogen, 3% – green hydrogen.

Steam methane reforming is the main process to produce blue and grey hydrogen [5,6]. Even though the first studies on steam methane reforming (SMR) were carried out more than a century ago, this topic is still relevant today. The investigations are mainly focused on the

development of new catalysts [7–9] and the design of new reformers for efficient hydrogen production [10,11]. The variety of studies on reformer designs is associated primarily with complex processes occurring in the reaction space: endothermic reactions, heat and mass transfer, diffusion, etc [12–14]. Moreover, SMR is the catalytic process and the catalysts are presented as a porous medium [15]. Therefore, the transport processes in the porous medium are also making the investigation more difficult.

To predict the parameters of the SMR process such as reformat composition, heat-mass flows, temperature distribution, etc, some approaches were proposed. The analytical calculations based on a conjugate solution of the mass, energy, and momentum conservation equations, and the chemical kinetic equations can be performed only for simple geometries such as plate, single tube, etc [16,17]. If the more realistic geometries of the methane reformer are considered, numerical simulations can be applied. Various software for numerical simulation can be used for that goal: Ansys Fluent, Comsol, FlowVision, Mathad, etc [18,19].

Computation fluid dynamics (CFD) modeling is most common approach for the numerical simulations of SMR [20–22]. A set of papers

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by Dixon and his co-authors significantly contributed to the problem of CFD modeling of the steam methane reforming process [23–25]. Dixon [26] presented the results of CFD-modeling of the SMR process in a packed bed randomly filled with 807 catalysts. The heat and mass transfer process between the reaction mixture and catalyst particles were simulated with diffusion, conduction, and reaction inside the catalyst particles. Dixon considered constant heat flux through a wall. The computation time for this simulation was about 30 days. This fact shows that for numerical simulation of the industrial-scale methane reformers, a particle-resolved approach will meet with huge computational time.

The performances of the SMR process depend not only on operational parameters such as temperature, pressure, steam-to-methane ratio, etc. but also they depend on the shape of the catalyst particles. Karthik and Buwa [27] studied the effect of catalyst particle shapes on fluid flow and heat transfer in a packed bed of a steam methane reformer. The reformer was filled with 10 catalyst particles. In addition, Karthik et al. [28] presented the results of particle-resolved modeling of a steam methane reformer to understand the effect of turbulence models on the results. The simulations were performed for the packed bed filled with 30 spherical catalyst particles. The authors concluded that an increase in a number of particles will notable increase in computational time.

Wehinger and Flaischlen [29] performed CFD-modeling of the SMR process to understand the effect of various radiation models on the heat and mass transfer. Usually, SMR operates in the temperature range from 500 to 1000 °C and radiation for a high-temperature process can play a significant role. The author used a particle-resolved approach for a packed bed with 588 particles. The calculation was performed via STAR-CCM+ software using 46 Intel Xeon 3.0 GHz processors. Even for such a powerful cluster, the calculation time for one step was about 5 min. Moreover, in this paper, the authors considered only surface SMR reaction without intra-particle heat and mass transfer.

Kuroki et al. [30] also performed CFD-modeling of SMR where the chemical reactions were considered as the surface reactions. The packed bed of the methane reformer consists of 550 spherical particles. The intra-particle processes such as diffusion, conductivity, and chemical reactions were not considered.

Many scientists and engineers have presented the novel results in numerical simulation of steam methane reforming via particle-resolved computational fluid dynamics. In 2020, Dixon et al. [31] presented a review where various aspects of the simulation of the packed tubular reactors via CFD were considered and analyzed. Moreover, Jurtz and co-authors [32] published the review paper “Advances in fixed-bed reactor modeling using particle-resolved computational fluid dynamics (CFD)”. In this paper, they showed significant progress in the understanding of the topic. However, the authors note that particle-resolved CFD requires significant computing power.

The discrete element method (DEM) can be applied for the numerical modeling of steam methane reforming [33]. DEM is a tool to model many discrete particles that connect with other particles and the gas flow. Using a DEM approach requires significantly less computing power than a particle-resolved simulation. Lu and Nikrityuk [34] published a new DEM-based model for numerical simulation of steam methane reforming. The calculations were performed for a packed bed filled with 807 spherical particles as in Dixon’s paper [26]. The results obtained via the DEM-based model were in good agreement with data from a particle-resolved model (3% deviation). Despite the advantages of this method, a packed bed is presented as many particles while the realistic industrial reformers contain thousands of particles.

Tran et al. [35] proposed to use the pseudo-homogeneous reactor model in which a packed bed is modeled by the continuum approach. In their paper, a packed bed presented as a porous zone function based on Ergun equation with the permeability and inertial loss coefficients [36]. This approach made it possible to model an industrial scale reformer with 336 reactor tubes. There is a set of papers with the criticism of

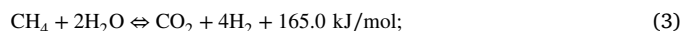
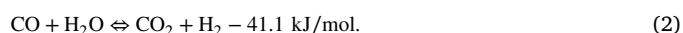
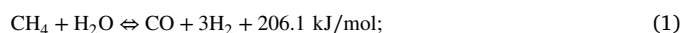
Ergun equation to predict the performances of the packed beds. However, the modern software allows to manually set the parameters of the porous medium via the permeability and inertial loss coefficients. The results of paper [35] showed the possibility of CFD-modeling of SMR in the industrial scale reformers with considering the in-particle processes. Tran et al. proposed to use the effectiveness factor in the equations from the chemical kinetic model. In their paper, the effectiveness factor as a constant ($\eta = 0.1$).

The main idea of the application of the effectiveness factor is to take into account the intra-particle processes, in particular, diffusional limitation. A catalyst particle cannot be fully used in the steam methane reforming process because a pore structure in a catalyst particle reduces the mass transport. Moreover, the reaction rates of SMR at the area near to catalyst core are decreasing because the species mole fraction decreases while the reformate mole fractions increase in the interior of the catalyst particles. The effectiveness factor is a non-constant value that depends on various parameters: the size of the catalyst particle, pore structure, temperature, mole fraction, etc. Based on a literature review, it was established that for a direct steam methane reforming reaction, the effectiveness factor varies from 0.001 to 1. Xu and Froment proposed that for the catalysts particles up to 0.2 mm, diffusion limitation is negligible and the effectiveness factor can be considered as unity [37]. However, Xu and Froment [38] in their next paper showed that for the catalyst particles of 1–1.2 mm, the effectiveness factor (η) is less than 1, while for a commercial steam methane reformer this factor is less than 0.03. Other articles also show that the factor η is influenced by particle size, temperature, the concentration of components, etc. [39–41].

The use of the pseudo-homogeneous reactor model in which a packed bed is modeled as a porous medium seems a promising way to predict the performances of the industrial-scale reformer. However, without accurate accounting of the processes inside the catalyst particles, the results will be weak. The effectiveness factor makes it possible to take into account these processes via addition to the equations of the chemical kinetic model. In this paper, the effect of various operation and design parameters (catalyst size and shape, porous medium structure, temperature and concentration, gas flow velocity) on the effectiveness factor is determined. The new approach of the pseudo-homogeneous SMR reactor model, in which the effectiveness factor is non-constant, is proposed and experimentally verified.

2. Methodology

Steam methane reforming is a well-investigated catalytic process. Various chemical reactions are taking place in the catalyst particles, including reactions of steam methane reforming, water-gas shift, carbon formation, and carbon gasification, etc. However, there is sufficient information in the literature that only three reactions have a notable effect on the energy and mass balances [37,42]:



During the last few decades, numerous kinetic models for the steam methane reforming (SMR) process have been proposed. The two most commonly cited models, as per the Scopus citation database, are the models developed by Xu and Froment for a Ni-based catalyst on MgAl_2O_4 [37] and Hou and Hughes for a Ni/ $\alpha\text{-Al}_2\text{O}_3$ catalyst [43]. Both models were derived using the Langmuir–Hinshelwood–Hougen–Watson (LH-HW) technique, and the chemical kinetic model equations have similar forms. In this study, the model provided by Hou and Hughes was utilized because the composition of their catalyst ($\text{NiO} = 15\%–17\%$, support Al_2O_3) and properties in their experiments are similar to those of a wide range of Ni-based industrial catalysts.

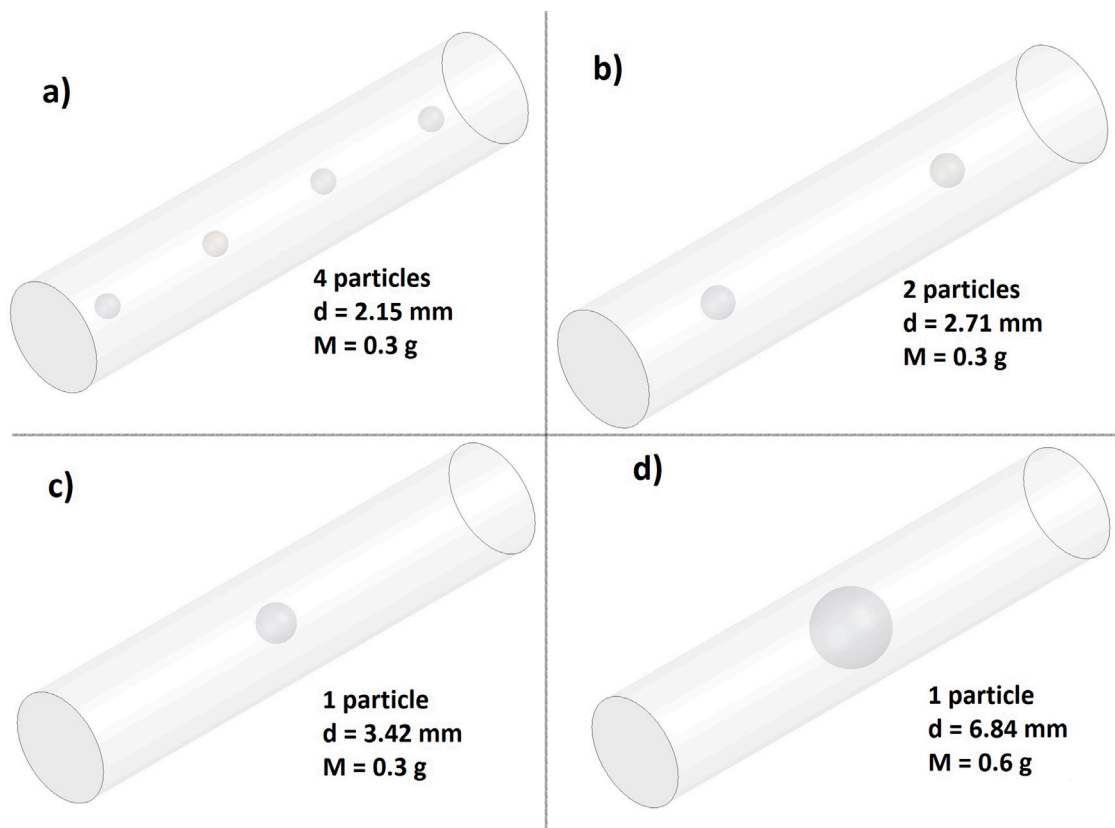


Fig. 1. The computational domains for the reaction space with the catalyst particles.

Hou and Hughes [43] showed that the steam methane reforming process is operating without diffusion limitation for the catalysts particles of 120–180 μm (0.12–0.18 mm). Their experiments for a model of intrinsic kinetics were performed in a tubular reformer of 10 mm in diameter and 338 mm in length. Moreover, the authors used 0.3 g of a Ni-based catalyst that crashed into 0.12–0.18 mm particles. Obviously, 0.3 g of catalyst is not enough to fully fill the reaction space of the reformer. Therefore, the catalyst particles were diluted with inert $\alpha\text{-Al}_2\text{O}_3$. The reformer length of 338 mm made it possible to maintain temperature along the reaction space close to constant.

The procedure of this current investigation is the following. In the first stage, the CFD model was developed to simulate the SMR process without diffusion limitation based on the intrinsic kinetic model developed by Hou and Hughes. The results were verified with the experimental data. In the second stage, the CFD model with diffusion limitation inside the catalyst particles was developed. In this model, catalyst particles of various sizes were tested under various operating conditions to understand the effect of the particle size and operating parameters on diffusion limitation and reforming performances.

2.1. Geometry and mesh

The geometry of a reformer is fully repeating a reformer for the experiments provided by Hou and Hughes [43]. The reaction space is 338 mm in length with side sections of 215 mm. A diameter is 10 mm for a reformer and side sections. The side sections are used to get a developed flow before the reaction space and to avoid a reverse flow at a reformer outlet.

A set of calculations was performed for 0.3 g of catalyst for various particle diameters. The catalyst particles are modeled as a single porous media. The number of particles is set to get 0.3 g of catalyst as presented in Table 1.

Table 1
Catalyst particle diameters.

Number particles	Diameter, mm	Total mass, g
64	0.85	0.3
32	1.08	0.3
16	1.36	0.3
8	1.71	0.3
4	2.15	0.3
2	2.71	0.3
1	3.42	0.3

The particles are uniformly distributed in the reaction space as shown in Fig. 1. Fig. 1 shows 4 cases: (a), (b), (c) is from Table 1; (d) shows a case that a catalyst particle is two times higher than for the case with one catalyst particle. In case (d), the total mass of the catalyst is 0.6 g and the diameter is 6.84 mm.

Despite the computational domain is simple, the special attention is paid to the mesh and its structure. In this paper, intra-particle processes are analyzed. Therefore, special attention is paid to the mesh structure of the catalyst particles. Fig. 2 depicts the mesh structure for intra-particle computational zone. The mesh structure for an intra-particle zone is the same for all cases from Table 1. Therefore, mesh independence for flow zone outside of catalyst particles is omitted in this study because the main efforts are made to uniform the mesh structure.

A mesh independence study was performed for the intra-particle domain using various numbers of mesh elements. Additionally, the mesh structure for the flow zone outside of the catalyst particle was tested using different numbers of elements. The intra-particle mesh structure and the mesh structure of the flow zone outside of the particle remained the same for all computation domains. The discretization strategy for the flow zone outside of the particle was chosen from [44]. Table 2 shows the effect of the number of elements in the outside

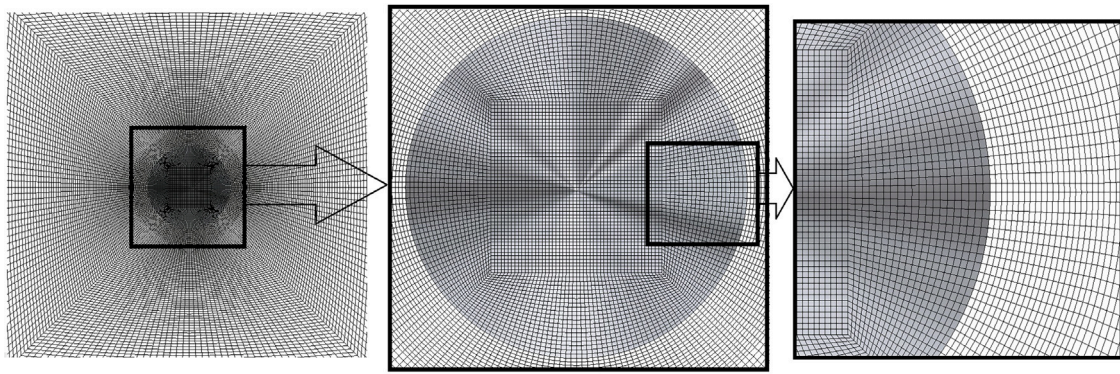


Fig. 2. The mesh structure for intra-particle computational zone.

Table 2

Influence of mesh elements of outside particle flow zone and intra-particle zone on drag coefficient and methane conversion.

Outside particle		Intra-particle	
Number of elements	C_d	Number of elements	CH ₄ conversion
$1.8 \cdot 10^5$	1.095	$8.7 \cdot 10^4$	0.0376
$9.7 \cdot 10^5$	1.091	$1.9 \cdot 10^5$	0.0332
$3.3 \cdot 10^6$	1.090	$7.1 \cdot 10^5$	0.0328
$1.3 \cdot 10^7$	1.090	$1.7 \cdot 10^6$	0.0327

particle flow domain and intra-particle domain on drag coefficient and methane conversion, respectively. The number of elements in the intra-particle zone was tested for a number of elements in the outside particle zone of $1.3 \cdot 10^7$. The data in Table 2 was obtained for a catalyst particle diameter of 3.42 mm.

2.2. Governing equations

The computational domain consists of two types of flow zones: without chemical reactions and with chemical reactions. Therefore, two forms of Navies–Stokes equations are used. The continuity equation for both flows (non-reaction and reaction) steady flow is the following:

$$\nabla \cdot (\rho \vec{v}) = 0 \quad (4)$$

The momentum equation has the difference because the reaction zone consists of the particles and an additional term should be involved in that equation:

$$\nabla \cdot (\rho \vec{v} \vec{v}) = -\nabla \cdot p + \nabla \cdot (\vec{\tau}) + \rho \vec{g} + \vec{F} \quad (5)$$

Porous catalyst particles and inert Al₂O particles are modeled by the addition of a momentum source term (S_i) to Eq. (5). The source term is composed of two parts: a viscous loss term (first term in eq.) and an inertial loss term. There are various models to describe the term (S_i) for a packed bed. One of them is the Ergun equation for the packed beds. Despite many criticisms, Ergun's approach shows an appropriate prediction for the high “tube diameter”/“particle diameter” ratio [45–47]. In the present case, this ratio is more than 50. Therefore, the expression for a momentum source term S_i has the following form:

$$S_i = - \left(\frac{150\mu(1 - \epsilon_{ip})^2}{D_p^2 \epsilon_{ip}^3} v_i + \frac{D_p^2}{150} \frac{\epsilon_{ip}^3}{(1 - \epsilon_{ip})^2} \frac{1}{2} \rho |v| v_i \right) \quad (6)$$

where μ is the viscosity; D_p is the mean diameter of pore of the catalyst and inert particles; ϵ_{ip} is the packed bed porosity.

For the model with diffusion limitation, the porous catalyst particles are separately considered. The actual pore size is 10–100 nm. There is some criticism of the use of porous media description of flow inside the catalyst particles. Dixon et al. [48] reported that the porous media model inside catalyst pellets coupled with the surrounding flow field is

erroneous and leads to wrong model predictions. Since the necessary no-slip condition at the gas–solid interface is violated, momentum is transported into the pellet influencing the local temperature and concentration distribution. However, the pore size of about 10–100 nm while the diameter of the catalyst particles for the industrial reformers can be up to 20 mm. Therefore, the momentum that is transported into the catalyst has negligible influence on the local temperature and concentration distribution for the whole catalyst particle. Moreover, the flow in the porous media is laminar and the source term (S_i) is defined by a permeability (α):

$$S_i = -\frac{\mu}{\alpha} v_i \quad (7)$$

A permeability (α) depends on pore size, tortuosity, etc. Therefore, experimental data for permeability from literature can be used in the CFD model for the given catalysts.

The reactions (Eq. (1)–(3)) are occurring in the volume of catalyst particles for the case with diffusion limitation and in the packed bed for the case of intrinsic kinetic (without diffusion limitation). Transport of j component can be determined by the following equation:

$$\nabla \cdot (\rho \cdot \vec{v} \cdot Y_j) = -\nabla \cdot \vec{J}_j + R_j \quad (8)$$

where R_j is reaction rate of formation or consumption of j th element; Y_j is mass fraction of j th element in the gas mixture; $\vec{J}_j$ is diffusion flux of j element.

For the surrounding flow of the catalyst particles, there are no reactions; therefore, R_j is zero. And the diffusion processes of j th component in the surrounding flow can be expressed as:

$$\vec{J}_j = - \left(\rho \cdot D_{j,m} + \frac{\mu_t}{Sc_t} \nabla Y_j - D_{T,j} \frac{\nabla T}{T} \right) \quad (9)$$

where $D_{j,m}$ and $D_{T,j}$ are the mass diffusion coefficient and thermal diffusion coefficient for j species, respectively; Sc_t and μ_t are the turbulent Schmidt number and turbulent viscosity, respectively.

For the intra-particle volume, the transport process is determined by in-particle diffusion and chemical reactions. The structure of the porous particles is very complex because it consists of many interconnected pores with various forms, various diameters, etc. Xu and Froment proposed to use the following expression for an effective diffusion inside the catalyst particles for j th element:

$$D_{eff,j} = \frac{\epsilon}{\tau} \bar{D}_j, \quad (10)$$

where τ is tortuosity factor; ϵ is porosity of catalyst particle; $\bar{D}_j$ is diffusion coefficient of j element.

The diffusion coefficient D_j is obtained by using Lewis number equal to 1.

$$Le = \frac{k_g}{D_j c_p \rho} = 1; D_j = \frac{k_g}{c_p \cdot \rho}. \quad (11)$$

Table 3
The coefficients for the equations of kinetic model.

Reaction rate		
k_1	=	$5.922 \cdot 10^8 \exp\left(\frac{-209200}{R_g T}\right)$
k_2	=	$6.028 \cdot 10^{-4} \exp\left(\frac{-15400}{R_g T}\right)$
k_3	=	$1.093 \cdot 10^3 \exp\left(\frac{-109400}{R_g T}\right)$
Adsorption		
K_{H_2O}	=	$9.251 \cdot \exp\left(\frac{-15900}{R_g T}\right)$
K_{H_2}	=	$5.68 \cdot 10^{-10} \exp\left(\frac{93400}{R_g T}\right)$
K_{CO}	=	$5.127 \times 10^{-13} \exp\left(\frac{140000}{R_g T}\right)$
Equilibrium		
K_1	=	$1.198 \cdot 10^{17} \exp\left(\frac{-26830}{T}\right)$
K_2	=	$1.767 \cdot 10^{-2} \exp\left(\frac{4400}{T}\right)$
K_3	=	$2.117 \cdot 10^{15} \exp\left(\frac{-22430}{T}\right)$

where transport properties k_g , c_p and density ρ are functions of temperature and are determined using polynomials taken from the database [49].

A tortuosity factor in Eq. (10) can be varied from 1.5 to 8 for the different catalyst supports. There are no appropriate equations to calculate tortuosity; therefore, this parameter can be estimated experimentally. Wakao and Smith proposed to calculate tortuosity as $\tau = 1/\epsilon$ [50]. Various experimental investigations showed a good agreement with this expression. For example, tortuosity is from 2 to 2.5 for α -Al₂O support.

The reaction rates of the chemical reactions (1)–(3) are determined based on the kinetic model presented by Hou and Hughes [43]:

$$R_1 = \frac{k_1(p_{CH_4}p_{H_2O}^{0.5} - p_{H_2}^3p_{CO}/K_1p_{H_2O}^{0.5})}{p_{H_2}^{1.25} \cdot DEN^2} \quad (12)$$

$$R_2 = \frac{k_2(p_{CO}p_{H_2O}^{0.5} - p_{H_2}p_{CO_2}/K_2p_{H_2O}^{0.5})}{p_{H_2}^{0.5} \cdot DEN^2} \quad (13)$$

$$R_3 = \frac{k_3(p_{CH_4}p_{H_2O} - p_{H_2}^4p_{CO_2}/K_3p_{H_2O})}{p_{H_2}^{3.5} \cdot DEN^2} \quad (14)$$

$$DEN = 1 + K_{CO} \cdot p_{CO} + K_H \cdot p_{H_2}^{0.5} + K_{H_2O} \cdot p_{H_2O}/p_{H_2} \quad (15)$$

The expression for the coefficients in (12)–(14) are presented in Table 3:

The expressions (12)–(14) are working only for the chemical reaction without diffusion limitation in the catalyst particles (intrinsic kinetic). For the catalyst particles with a diameter above 0.20–0.25 mm intra-particle diffusion is taking a place. To take into account the diffusion limitation, an effectiveness factor should be added to the expression (12)–(14). The effectiveness factor can be determined as follows:

$$\eta = \frac{R_{real,j}}{R_{intr,j}} \quad (16)$$

where $R_{real,j}$ is reaction rate with diffusion limitation; $R_{intr,j}$ is reaction rate without diffusion limitation (intrinsic kinetic).

One of the purposes of this study is to determine the effect of operating parameters and properties of a catalyst on the effectiveness factor. A computational algorithm of the effectiveness factor is presented below.

The attempts at the analytical calculation of the effectiveness factor via the Thiele module were described in a set of publications [51–53]. Moreover, Baek et al. [53] expressed the effectiveness factor as a function of the methane conversion. However, the use of both approaches for numerical modeling via modern CFD tools is a difficult task because in those cases UDF for chemical kinetics will not be able to compile. Therefore, the representation of the effectiveness factor as a simple function of operational and design parameters is an important task.

2.3. Boundary conditions

A set of numerical calculations was performed for a wide range of operating parameters such as temperature, steam-to-methane ratio, and residence time (set via the mass flow rate of the reaction mixture). The Reynolds number is varied from 500 to 2000 for various residence time. Moreover, various pressures were tested to understand their effect on the diffusion limitation. As it was mentioned above, the calculations were performed for the intrinsic kinetic model (catalyst particle of 0.15 mm) and for the model with diffusion limitation. For the second case, the mass of the catalyst of 0.3 g was presented via the catalyst particles with the diameters presented in Table 1.

The catalyst particles from Table 1 are presented as single porous spheres with the following parameter: porosity – 0.39; Darcy permeability – $2.8 \cdot 10^{-14}$ m²; Forchheimer permeability – $8.4 \cdot 10^{-9}$ m. Although the fluid flow in a porous media follows Darcy's Law (linear rate law) in practical cases, at flows the Forchheimer (nonlinear) flow regime emerges, where fluid flux and the hydraulic gradient are no longer linearly related. Therefore, the Forchheimer permeability is also taken into account.

The temperature in the reaction space for the experiments performed by Hou and Hughes was close to constant due to the heat supply through the wall. In the numerical calculations, the temperature of the reformer wall is set equal to the temperature of the reaction mixture for each calculation. The SMR process is highly endothermic and there are temperature gradients in the reaction space. However, due to the large length of the reformer, the temperature of the reaction mixture is stabilized by the heat supply.

A mathematical formulation of the boundary conditions can be expressed as the following:

$$\begin{aligned} &\text{– reformer inlet: } T_{in} = T_{in}^{gas}; Y_{i,in} = Y_{i,in}^{gas}; \\ &\text{– reformer outlet: } \frac{\partial Y_i}{\partial x} = 0; \frac{\partial T^{gas}}{\partial x} = 0. \end{aligned}$$

In the paper presented by Hou and Hughes [43], the residence time (time of contact between reaction mixture and catalyst) was varied up to 14 000 kg_{cat} s/mol_{CH₄}. In the present paper, the residence time is varied via changes in the mass flow rate of the reaction mixture.

3. Results and discussion

3.1. Intrinsic kinetics

As noted above, the kinetic model suggested by Hou and Hughes [43] was used to determine the rate of formation and consumption of the elements in the reaction space. To calculate the reaction rates, the CFD model with intrinsic kinetics was developed. In this model, the reaction space filled with the catalyst particles without diffusion limitation (size of particles is 0.15 mm) set as a porous medium. The reaction space is filled with the catalyst and inert particles. To take into account the fact that only the catalytic part of reaction space is playing role in the reforming process, the effectiveness factor (η_{int}) was introduced to the kinetic model:

$$\eta_{int} = \frac{M_{cat}}{M_{total}} \quad (17)$$

where M_{cat} is mass of the catalyst particles; M_{total} is total mass of the active and inert particles.

According to Hou's paper, the mass of the catalyst is 0.3 g and the diameter of the particle is 0.15 mm. The volume of reaction space is formed by a tube with a length of 338 mm and a diameter of 10 mm. The inert particles consist of the same Al₂O support. Therefore, it was assumed that the density of active and inert particles is the same.

The main goal of the calculation of steam methane reforming with intrinsic kinetics is to obtain the values of reaction rates for the case of no diffusion limitation. Fig. 3a shows the comparison of the numerical results obtained in this paper with the experimental results obtained by Hou and Hughes [43]. The numerical results were obtained for p

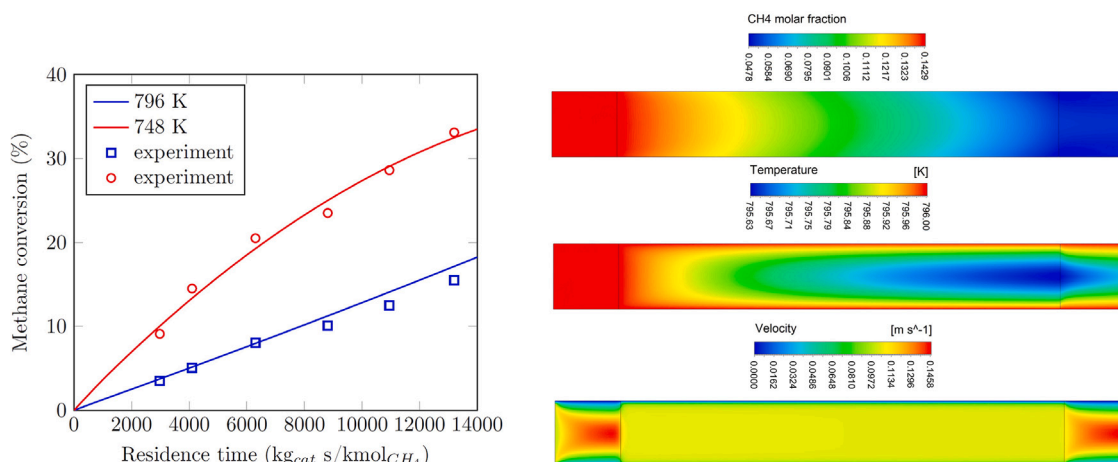


Fig. 3. (a) CH₄ conversion vs. residence time: solid lines — CFD modeling, hollow points — experimental results [43]; (b) Contours of CH₄ mole fraction, temperature and velocity at 796 K, $p = 300$ kPa, CH₄:H₂O:H₂ = 1:5:1, residence time 12000 kg_{cat} s/kmol_{CH₄}.

= 300 kPa, CH₄:H₂O:H₂ = 1 : 5 : 1. The residence time was varied through the change in mass flow rate of the inlet mixture. Fig. 3a depicts a good correlation between the numerical and experimental results.

Moreover, Fig. 3b presents the contours of CH₄ mole fraction, temperature and velocity at 796 K, $p = 300$ kPa, CH₄:H₂O:H₂ = 1 : 5 : 1, residence time 12000 kg_{cat} s/kmol_{CH₄}. The distribution of these parameters is shown for three zones: inlet zone, reaction space, and outlet zone. To better understand the distribution of the parameters in the reformer with the design parameters from Hou's paper, scale in a radial direction was increased for all zones, and decreased for an axial direction for the inlet and outlet zones. Fig. 3b shows the change in molar fraction not only in the axial direction but also in the radial direction. This fact can be explained by the velocity distribution in the reaction zone. A flow adhesion is observed for the near-wall region. The temperature distribution does not have a notable effect on CH₄ conversion because the temperature deviation inside the reaction zone is less than 1 K. This fact is in good agreement with Hou's paper [43] where the authors stated that the flow inside the reaction zone can be considered as isothermal flow.

The results from Fig. 3a show that the CFD model based on the kinetic model proposed by Hou and Hughes is appropriate for numerical modeling of the steam methane reforming process when diffusion limitation inside catalyst particles is omitted. To understand the diffusion limitation inside catalyst particles, a set of calculations was performed. The procedure for the numerical calculations is described in Section 2.1.

3.2. Diffusion limitation

To understand the effect of diffusion limitation, numerical calculations were performed for the same boundary and initial conditions. However, 0.3 g of catalyst was placed into the reformer as a particle. The reformer space is filled with inert Al₂O supporter as for the case with intrinsic kinetic in Section 3.1. Therefore, the porous catalyst particles were placed in the reformer space filled with inert Al₂O supporter. To simulate the heat and mass transfer processes in such system, two porous zones were marked in the computational domain. The first one (Al₂O supporter) is the non-reaction zone, and the second one is the reaction zone (catalyst particles).

Fig. 4 shows the contours of CH₄ and H₂ molar fractions, as well as the contours of velocity. The figure illustrates that no chemical reactions are observed in the reaction space filled with inert particles. Simultaneously, reactions are taking place inside the catalyst particles. The maximum methane conversion is observed in the particle cores

because mass transfer via in-particle diffusion is a slower process. Comparing the contours for CH₄ mole fraction from Figs. 3 and 4, it can be seen that in-particle diffusion decreases methane conversion. The quantitative comparison of CH₄ conversion from Figs. 3 and 4 shows that there are notable mass transfer limitations within the catalyst particles, and the total CH₄ conversion for the intrinsic kinetic case is about 20 times higher than that for the particle of 3.42 mm in diameter.

Moreover, Fig. 4 presents the contours of velocity for the reformer space. Fig. 4 shows that the velocity contour for the reformer space filled with the inert particles is almost the same as for the case of intrinsic kinetics. However, in the area near the catalyst particle, the velocity is increasing because the permeability coefficient of the layer filled with inert particles is notably higher than that for the catalyst particles. In addition, Fig. 4 (bottom) shows that no transport momentum is observed into the catalyst particle. Therefore, the porous media model inside catalyst pellets coupled with the surrounding flow field is appropriate and leads to good model predictions. Dixon et al. [48] proposed that the catalyst particles should be treated as solid particles in the CFD model. However, in the present paper, the flow around the catalyst particles is low-Reynolds and the velocity is less than 0.2 m/s. Therefore, no transport momentum inside particles is observed and mass transfer is led by diffusion.

The effect of particle diameter on the effectiveness factor is presented in Fig. 5. The curves show a non-linear relationship between the effectiveness factor and diameter of the catalyst particle for the reactions (12) and (14). Fig. 5 depicts the effectiveness factor for 900 K and 1200 K for a steam-to-methane ratio of 3.0. Fig. 5 illustrates that there is significant mass transfer limitation inside the catalyst particles when the diameter is increasing. For the intrinsic kinetics, the effectiveness factor will equal unity. The reactions are taking place within the inner volume of the catalyst particle. Therefore, the concentration of steam and methane is reduced and the concentration of hydrogen and carbon monoxide is increased inside the catalyst pellets. Diffusion and penetration of H₂ and CO to the catalyst surface as well as diffusion and penetration of CH₄ and H₂O from the surface to the particle core is strongly dependent on the distance from the catalyst surface to the catalyst core.

Moreover, Fig. 5 shows that the effectiveness factor for reactions (1) and (3) is a little lower than for reaction (1). The main reason is carbon dioxide produced by the reaction (3). The diffusion of CO₂ is lower than the diffusion of CO. The effectiveness factor for reaction 2 is constant and equal to close to 1. In the temperature range above 900 K, the water gas shift reaction can be considered as a near-equilibrium reaction. Therefore, the effect of reaction 2 on mass and heat balance is negligible.

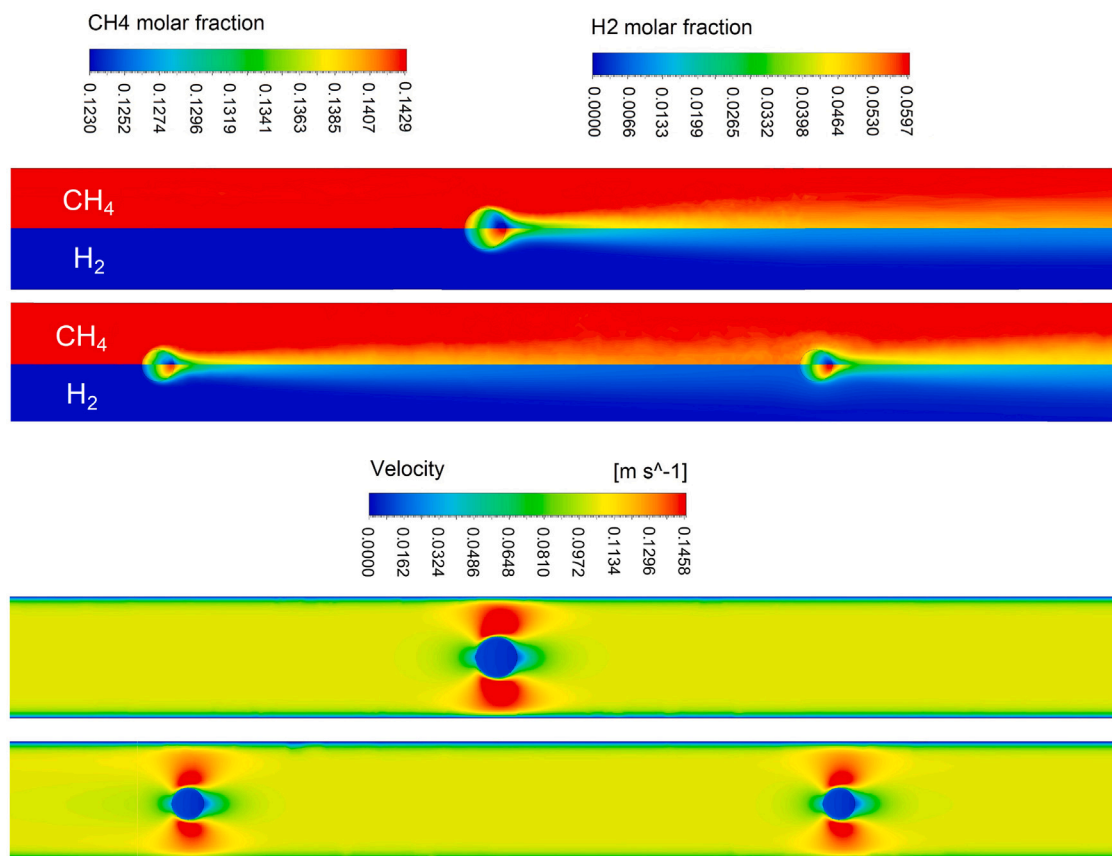


Fig. 4. Contours of CH₄ and H₂ mole fraction (top), and velocity (bottom) at 796 K, $p = 300$ kPa, CH₄:H₂O:H₂ = 1:5:1, residence time 12000 kg_{cat}/s/kmol_{CH₄}.

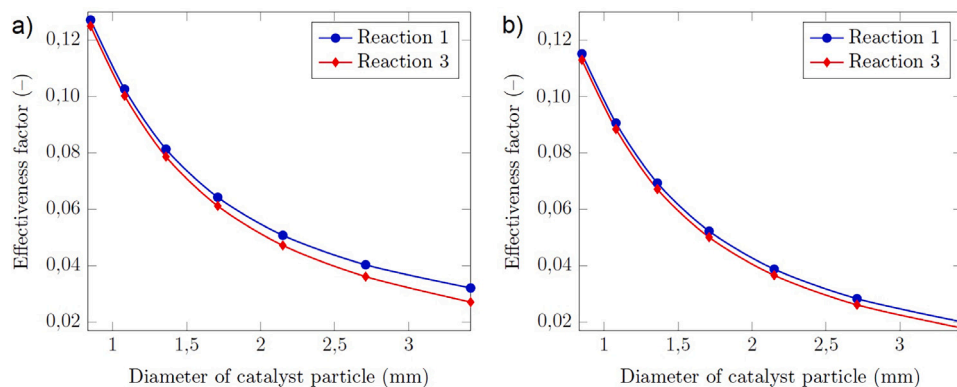


Fig. 5. Effectiveness factor as a function of catalyst particle diameter for Reaction (1) and (3) for CH₄:H₂O = 1:3, residence time 12000 kg_{cat}/s/kmol_{CH₄}: (a) $T = 900$ K, (b) $T = 1200$ K.

To assess the effect of operational parameters, the dependence of the effectiveness factor on temperature was determined as presented in Fig. 6. Fig. 6 shows that the effectiveness factor is slightly dependent on temperature. The effectiveness factor tends to decrease as the temperature of the process increases. The trend of the effectiveness factors for various particle diameters makes it clear that for the given temperature level the particle diameter has no negligible effect on the effectiveness factor. However, an increase in temperature leads to an increase in the rate of the effectiveness factor rate. In the temperature range of 800–900 K, the effectiveness factor can be constant while in the temperature range of 1200–1300 K, the change in the effectiveness factor is significant. This finding can be explained by the fact that an increase in the temperature leads to an increase in the reaction rate. In partial pressure of reforming products inside the catalyst particle is

increasing with an increase in temperature. Therefore, the effectiveness factor tends to decrease with an increasing reaction rate. Therefore, as a result, the dependencies from Fig. 6 can be interpreted as the effectiveness factor will tend to be lower when the reaction rate is higher. The effect of the temperature on the effectiveness factor for the reaction (3) is omitted because the trend lines are similar to the effectiveness factor for the reaction (1). Moreover, Fig. 6 shows that an increase in a steam-to-methane ratio decreases the effectiveness factor.

Fig. 7 depicts the effect of the steam-to-methane ratio on the effectiveness factor. An almost linear decrease of the effectiveness factor with a steam-to-methane ratio increase was observed from the figure at constant temperature and constant residence time. A possible reason for this observation is that an increase in steam-to-methane ratio leads to a decrease in CH₄ partial pressure. As a result, the methane diffusion

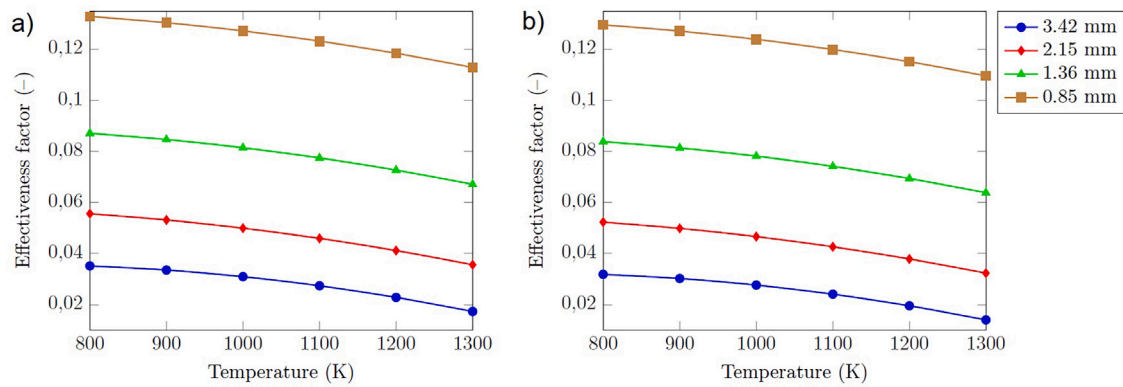


Fig. 6. Effectiveness factor as a function of temperature for Reaction (1) for residence time 12000 kg_{cat}/s/kmol_{CH₄} and T = 900 K: (a) CH₄:H₂O = 1:2, (b) CH₄:H₂O = 1:3.

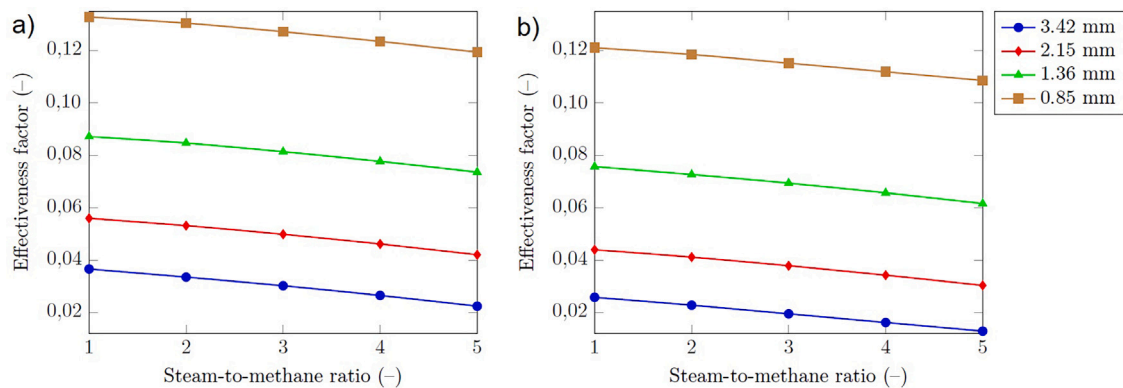


Fig. 7. Effectiveness factor as a function of a steam-to-methane ratio for Reaction (1) for residence time of 12000 kg_{cat}/s/kmol_{CH₄}: (a) T = 900 K, (b) T = 1200 K.

is decreasing faster than the reaction rate decreasing due to a decrease in CH₄ partial pressure. This fact can be confirmed by the following expressions [51]:

$$\eta_1 \approx 1/\phi_1; \quad \phi_1 = \sqrt{\frac{R_1}{M_{CH_4}}} \quad (18)$$

where ϕ – the Thiele modulus; R_1 – the nominal reaction rate of reaction (1); M_{CH_4} – the methane diffusion rate.

Fig. 7 shows that the effectiveness factor drops as temperatures increase. The effect of methane partial pressure on the effectiveness factor repeats the dependencies from Fig. 7 when the total pressure in the reformer is constant. The change of CH₄ partial pressure as a constant steam-to-methane ratio has an insignificant effect on the effectiveness factor in the analyzed total pressure range from 1 bar to 20 bar.

In addition to the presented dependencies, the effect of the residence time on the effectiveness factor was analyzed. The residence time was varied via the mass flow rate of the reaction mixture. However, no significant effect of the residence time was found in the range of 3000...15000 kg_{cat}/s/kmol_{CH₄}. This may be because the catalyst particles are placed into the packed bed filled with small particles of catalyst supporters (0.20 mm). Therefore, when the reactants move through such a bed, the velocity decreases markedly.

The dependencies from Figs. 5, 6 and 7 obtained not from a theoretical basis. They are just numerical variables to describe the effectiveness factor as a function of the main operational parameters such as temperature and steam-to-methane ratio as well as the size of the catalyst particles. The effectiveness factor is obtained for the basic operating conditions for the steam methane reforming process. Moreover, the dependencies from Figs. 5, 6 and 7 showed that the particle diameter gives the most notable effect on the effectiveness factor while temperature and steam-to-methane ratio gives a slight effect.

Based on the analysis, the equation for the calculation of the effectiveness factor as a function of the catalyst particle size, temperature and steam-to-methane ratio was obtained in the following form:

$$\eta = f_d(d) \cdot f_T\left(\frac{T}{1000}\right) \cdot f_b\left(\frac{N_{H_2O}}{N_{CH_4}}\right) \quad (19)$$

where $f_d(d)$ – the function of the effectiveness factor on diameter; $f_T\left(\frac{T}{1000}\right)$ – the function of the effectiveness factor on temperature; $f_b\left(\frac{N_{H_2O}}{N_{CH_4}}\right)$ – the function of the effectiveness factor on steam-to-methane ratio.

To determine the form-factor function $f_d(d)$, the conclusion from the following papers was used. Pedernera et al. [54] determined that steam methane reforming takes place in a zone close to the particle surface. They found that just about 2.5% of the total depth of the pellets is being used for the process. The authors analyzed the catalyst particle with a characteristic dimension of 11 mm. This fact confirmed the results from Fig. 5 where the effectiveness factor is increasing with a decrease in the diameter of the catalyst particle. For the catalyst particles with higher diameters, the notable part of the catalyst volume in a core does not affect the steam methane reforming process.

In addition, Xu and Froment [38] considered the catalyst particles for the numerical calculations with an active catalyst layer of 2 mm while the total size of the catalyst particles is 22 mm. This can be explained by the fact that only the zone near the catalyst surface plays an important role in the steam methane reforming process. Based on those publications, it can be expected that the effectiveness factor depends on the catalyst surface. For assessment of the effect of the catalyst surface on the effectiveness factor, the catalyst surface as a function of a catalyst diameter was determined. Fig. 8a shows the comparison of the dependencies for the catalyst surface and effectiveness factor

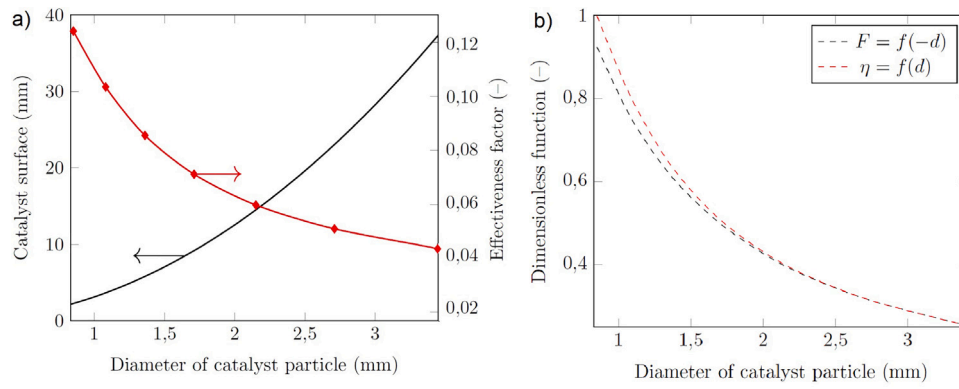


Fig. 8. Catalyst surface and effectiveness factor as a functions of the catalyst diameter.

on the catalyst diameter. Fig. 8a makes it clear that the nature of the dependence of the effectiveness factor and surface area of the catalyst particle on the diameter is symmetrical with respect to the abscissa axis. Therefore, $\eta = f(d)$ is similar to $F = f(-d)$. Based on this assumption, Fig. 8b was presented. Fig. 8b depicts the dependencies of the dimensionless function on the catalyst surface F . Fig. 8b illustrates that for the catalyst particles with $d > 2$ mm the effectiveness factor is proportional to the area of the catalyst surface. When the diameter decreases (less than $d < 2$ mm) the deviation is observed. This means that the active layer of the catalyst particle is about 1 mm and when the diameter is less than 2 mm, the diffusion limitation inside the catalyst particle is unproportional to the catalyst surface.

Taking into account the obtained results, the function $f_d(d)$, f_T , f_b from Eq. (19) were obtained using regression analysis. For the form-factor function $f_d(d)$ it was found a simple and accurate correlation:

$$f_d(d) = 0.11/d; \quad (20)$$

$$f_T\left(\frac{T}{1000}\right) = -1.38875 \cdot \left(\frac{T}{1000}\right)^2 + 1.90065 \cdot \left(\frac{T}{1000}\right) + 0.368183; \quad (21)$$

$$f_b\left(\frac{N_{H_2O}}{N_{CH_4}}\right) = 1.10535 - 0.0966 \cdot \left(\frac{N_{H_2O}}{N_{CH_4}}\right) \quad (22)$$

In the functions Eq. (20)–(22) unit for d is mm, for T is K, for N_i is mole. In this paper, various diameters of the catalyst particles were analyzed. 5 and 6 decimal places are used to provide 95% confidence intervals for the big particles ($d > 2$ mm). 3 and 4 decimal places are enough to get 95% for the small particles ($d < 2$ mm). The statistical analysis was performed to get each parameter.

The function $f_d(d)$ is appropriate for the spherical catalyst. It can be expected that the dependencies from Fig. 8b can be similar to other catalyst shapes. Therefore, the first term in Eq. (19) can be expressed as a function of a surface of catalyst particle (f_F).

$$f_F = 0.19/F^{0.5}. \quad (23)$$

where F – the geometrical (external) surface area of the catalyst particle (mm²).

Based on the expressions (20)–(23) and Eq. (19), the diffusion limitation expressed via the effectiveness factor can be determined for the industrial Ni-based catalyst.

As it was mentioned above, the current pseudo-homogeneous steam methane reformer models consider the effectiveness factor as constant for the whole reacting space [55]. However, the temperature deviation in the packed bed of the reformers can be up to 200 °C. Moreover, the reformer can be filled by the catalyst particles with various shapes. The above-obtained expressions can be used for the pseudo-homogeneous steam methane reformer models where the effect

of various parameters on the effectiveness factor is taken into account. Therefore, the presented analysis can be considered as a way to more accurate pseudo-homogeneous steam methane reformer models.

The main limitations of the present analysis are the following:

1. The permeability coefficient depends on a catalyst support. In this paper, $\alpha\text{-Al}_2\text{O}_3$ support is considered. It can be interesting to assess the effect of permeability on diffusion limitation. For this goal, the different catalyst supports should be tested.
2. Coking is another factor that needs to be considered especially at high temperatures. In this paper, the temperature range up to 1300 K was analyzed. Therefore, the chemical reactions of carbon formation were omitted. In real reformers, the carbon deposition will decrease the effectiveness factor.
3. Tortuosity and pore size are also playing an important role in diffusion limitation. In this paper, tortuosity and pore size is considered as a constant for all calculations.

4. Conclusion

In this work, the diffusion limitation within the catalyst particles for the steam methane reforming process was assessed. For this goal, the numerical calculations for the intrinsic kinetics of steam methane reforming as well as for the kinetics with diffusion limitation were performed via the commercial software Ansys Fluent. The CFD model was validated against the experimental results. Good agreement was achieved between the numerical and experimental results. The effect of the diffusion limitation was expressed via an effectiveness factor. It was established that an increase in the diameter of the spherical catalyst particles leads to a decrease in the effectiveness factor according to the function $\eta \approx 0.11/d$ (d is particle diameter, mm). For the catalyst particles with $d > 2$ mm the effectiveness factor is proportional to the area of the catalyst surface while for $d < 2$ mm the deviation is observed. Moreover, the effect of temperature and steam-to-methane ratio on the effectiveness factor was determined. An increase in temperature and steam-to-methane ratio leads to a decrease in the effectiveness factor. The correlation functions for the effect of temperature and steam-to-methane ratio on the effectiveness factor were expressed. The obtained results for the spherical particles were extrapolated for other particle shapes via an area of the catalyst surface. The effectiveness factor is depending on the area of the catalyst surface according to the following function: $\eta \approx 0.19/F^{0.5}$ (F is surface area, mm²). It was shown that the partial pressure of the reactants has a negligible effect on the effectiveness at a constant steam-to-methane ratio. The final expression for the effectiveness factor can be used for a pseudo-homogeneous steam methane reformer model where the effectiveness factor varies depending on the position in the packed bed.

CRediT authorship contribution statement

Dmitry Pashchenko: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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